**Table 16.2** Diseases caused by microorganisms in animals

| Animal diseases | Causative microorganisms | Mode of transmission | Symptoms | Preventive measures/Treatment |
|---|---|---|---|---|
| Anthrax (also affects humans) | *Bacillus anthracis* (Bacteria) | Through contaminated soil and food | Difficulty in breathing, unconsciousness, loss of appetite | Anthrax vaccine |
| Foot and mouth disease | *Aphthovirus* (Virus) | Through air and animal vectors | Fever, blisters in mouth, weight loss, decreased milk production | FMD vaccine |

**Table 16.3** Diseases caused by microorganisms in humans

| Human diseases | Causative microorganisms | Mode of transmission | Symptoms | Preventive measures/ Treatment |
|---|---|---|---|---|
| Tuberculosis | *Mycobacterium tuberculosis* (Bacteria) | Through air and sputum of infected person | Persistent cough, blood mucus, loss of weight, breathlessness | BCG Vaccine |
| Cholera | *Vibrio cholera* (Bacteria) | By flies and contaminated food and water | Watery diarrhoea, vomiting, rapid dehydration. | Anticholera vaccine, maintaining personal hygiene. |
| Common cold | *Influenza* (virus) | Through air | Running nose, sneezing | Isolation of patient |
| Rabies | *Rhabdo viridae* (Virus) | Animal bite | Fever, hallucination, paralysis, inability to swallow | Anti-rabies vaccine |
| Amoebic dysentery | *Entamoeba histolytica* (Protozoa) | Food water and flies | Severe diarrhea and blood in stool | Proper sanitation to be followed and metronidazole antibiotic to be administered |
| Malaria | *Plasmodium* (Protozoa) | Female anopheles mosquito | Nausea, vomiting, high fever | Antimalarial drugs like quinine and chloroquine to be taken and usage of mosquito repellents and nets. |

## 16.10 Relationship Between Man and Microbes- Balances, Imbalances and Uses

Thousands of bacteria, fungi and other microbes that live in our gut are essential contributors to a good health. They break down toxins, manufacture some vitamins and essential amino acids and form a barrier against invaders. Gut microbes are the bacteria in human gut. They are one of the most important allies in our overall health and well being. Gut ensures that the body is absorbing all the important nutrients, to function at its highest level. Many different aspects of health are attached to it.

## Points to Remember

- The organisms which can be seen only with the help of microscope are called microorganisms.



8th_Science_Unit-16_EM.indd 188 03-03-2020 15:37:55